Deep Learning Guide

Deep learning is a subset of machine learning that uses artificial neural networks containing multiple hidden layers.

Neural Networks

A neural network consists of:

Input layer

Hidden layers

Output layer

Each neuron learns numerical weights during training.

Convolutional Neural Networks

CNNs are widely used for image classification and computer vision.

Recurrent Neural Networks

RNNs process sequential information such as speech and time-series data.

Transformers

Transformers use the self-attention mechanism to model relationships between words.

Large Language Models such as GPT and BERT are based on Transformer architectures.